

AI FINANCIAL ADVICE: SUPPLY, DEMAND, AND LIFE CYCLE IMPLICATIONS

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Example: Gender gap in investment advice $\approx$ 2/3 demand, 1/3 supply

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State Variables

EXOGENOUS STATES

- Income process
- Employment transitions
- Asset returns
- Mortality risk

SIPP + SSA + CRSP

ENDOGENOUS STATES

- Wealth
- Asset allocation

State Variables



Policy Functions

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SIPP + SSA + CRSP

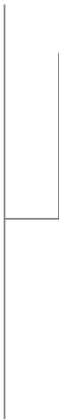
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BENCHMARK I

Life Cycle Model

Choukhmane-de Silva 2026



State Variables

Policy Functions

Simulated Choices

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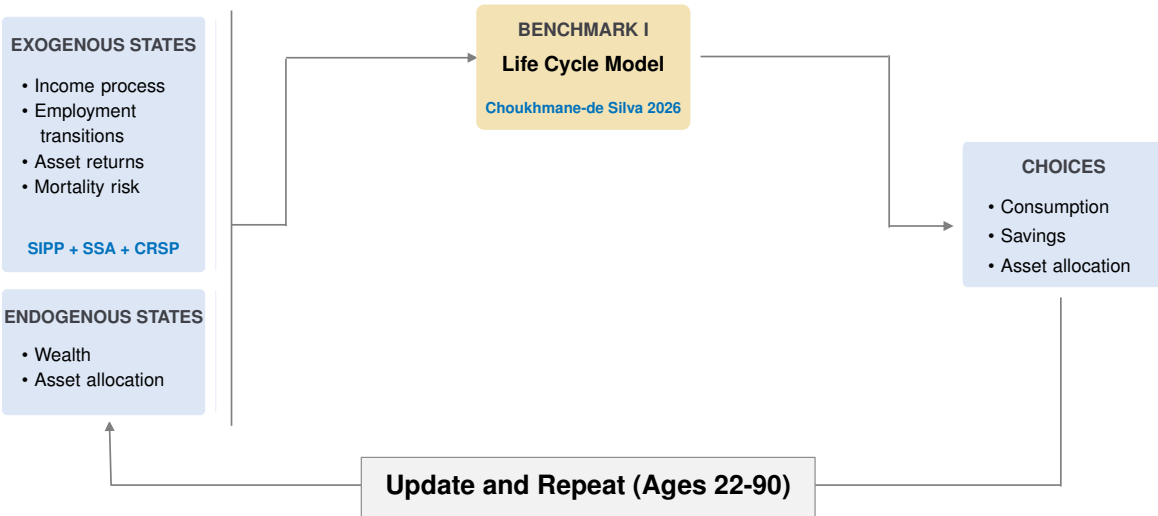
CHOICES

- Consumption
- Savings
- Asset allocation

State Variables

Policy Functions

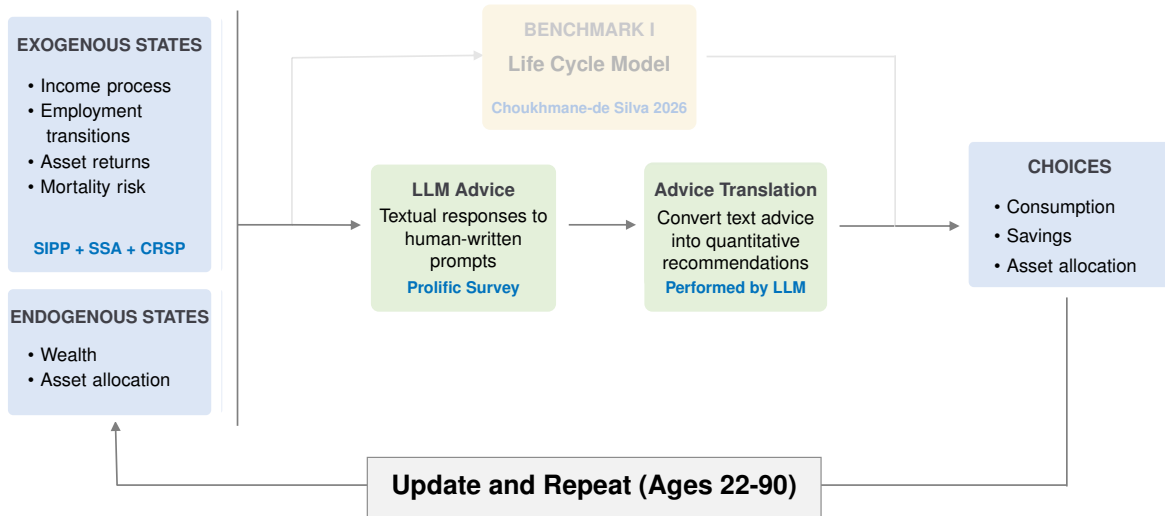
Simulated Choices



State Variables

Policy Functions

Simulated Choices



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PROMPT

ADVICE

CHOICES

SHOCKS

Simulated Individual

Age: 26

Employment status: Employed

Pre-tax income: \$68,810

Asset allocation:

- Fixed income: \$17,938
- Diversified stocks: \$6,442
- Individual stocks: \$0
- Alternative assets: \$0

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Prolific Prompt (Random by Age-Employment-Income)

Part 1: Financial Situation

I make around 75000 a year before taxes and pay 1700 a month in rent. That is my primary income and I make some money completing paid research surveys. I have around 80000 saved. I am 25.

Part 2: Spending Advice

I am looking for ways to go out more and enjoy my life without sacrificing my savings. I have a good grip on money and actually struggle to justify spending.

Part 3: Investing Advice

How should I put away money now that interest rates are going down. Safe bets like my money market and CDs are no longer yield much money.

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I make around {income} a year before taxes and pay 1700 a month in rent. That is my primary income and I make some money completing paid research surveys. I have around {wealth} saved. I am {age}.

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Prolific Prompt (Random by Age-Employment-Income)

Part 1: Financial Situation

I make around 68810 a year before taxes and pay 1700 a month in rent. That is my primary income and I make some money completing paid research surveys. I have around 24380 saved. I am 26.

Part 2: Spending Advice

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Prompt Sent to LLM (via API Call with Clear Context)

"I make around 68810 a year before taxes and pay 1700 a month in rent. That is my primary income and I make some money completing paid research surveys. I have around 24380 saved. I am 26.

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LLM Response

Given your income/savings, you can safely “permission” yourself to spend more without derailing goals by separating **spending from saving**.

- 1) **Set a floor, then automate:** Keep an emergency fund of **3-6 months** of essential expenses in a **HYSA/money market** (don’t chase yield; it’s for safety). Automate investing each payday (e.g., **15-25% of gross**, adjust to comfort). What’s left is guilt-free.
- 2) **Create a “fun” system:** Open a separate checking/savings called “FUN”. Auto-transfer a fixed amount (start **\$300-\$600/month**). If it’s there, you must use it or roll it to a bigger experience.
- 3) **Where to put money as rates fall:** For long-term (5+ years): **low-cost broad index funds/ETFs** (e.g., total US + total international) in 401(k)/Roth IRA first. For medium-term goals (1-5 years): HYSA + Treasuries/T-bill ladder (often tax-advantaged vs CDs), or a short-term bond fund if you can tolerate modest fluctuation.
- 4) **Psychology:** Treat spending from the FUN account as a goal, not a failure-your system already protected savings.

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LLM Advice (Translated with LLM)

Consumption: **\$44,378**
Total savings: **\$17,845**
Asset allocation:
- Fixed income: **+\$5,400**
- Diversified stocks: **+\$12,445**
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Labor Market Transitions

- Age and income specific job-to-job and unemployment transitions
- Persistent and transitory income shocks

Asset Returns

- Fixed income: constant risk-free rate
- Diversified stocks: CRSP Value-Weighted Index
- Individual stocks: CAPM + Bessembinder (2018)
- Alternative assets: individual stock return process with independent realizations

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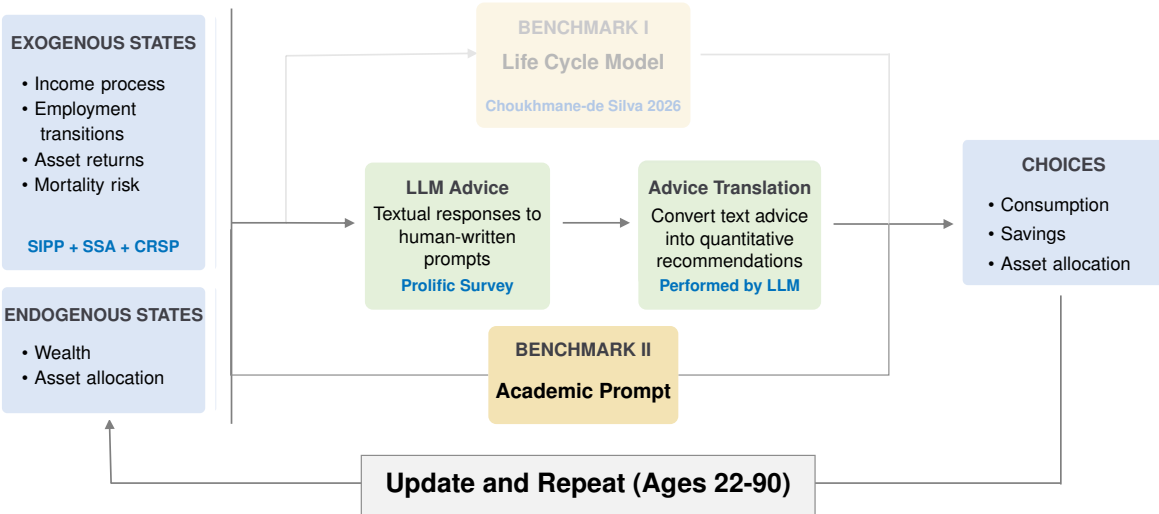
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Academic Prompt (Directly Outputs Numeric Choices)

You are a U.S. financial advisor acting in the best interest of your clients. You are academically trained in household finance, modern portfolio theory, and life cycle planning. **Your objective is to produce high-quality personalized financial advice tailored to my circumstances** under these baseline assumptions:

- Normal life expectancy, living expenditures, retirement age, employment risk, and income risk
- Current U.S. tax law and Social Security rules stay unchanged
- Risk-free savings earn 2% real return annually
- Real stock returns match the 60 year U.S. total stock market historical average
- I am single with no dependents and have no bequest motive. I do not care about wealth after death.

Given my ****age, employment status, annual post-tax salary, average annual post-tax income (since age 22), taxable account balances, and total net wealth****, give the best financial advice for allocating my resources across annual spending and the following four taxable accounts:

[...]

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 - ② *“In your own words, write a prompt asking an AI tool for advice on **how much to spend over the coming year**”*
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- Collect demographics, financial literacy, risk/time preferences, saving motives

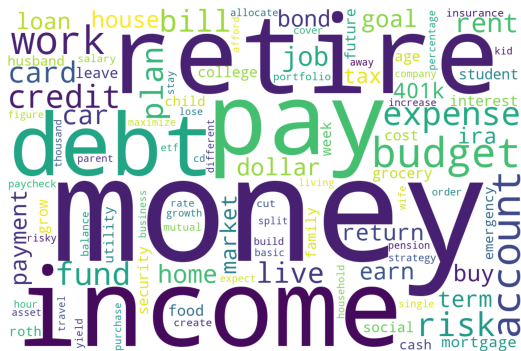
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- Collect demographics, financial literacy, risk/time preferences, saving motives
- Aim for authentic prompts: 952 pass quality check (AI detection + manual specificity)

*“Please write naturally, as you would if you were really asking advice for yourself. There are no right or wrong answers.
Please do not consult outside sources or use AI tools to complete this task”*

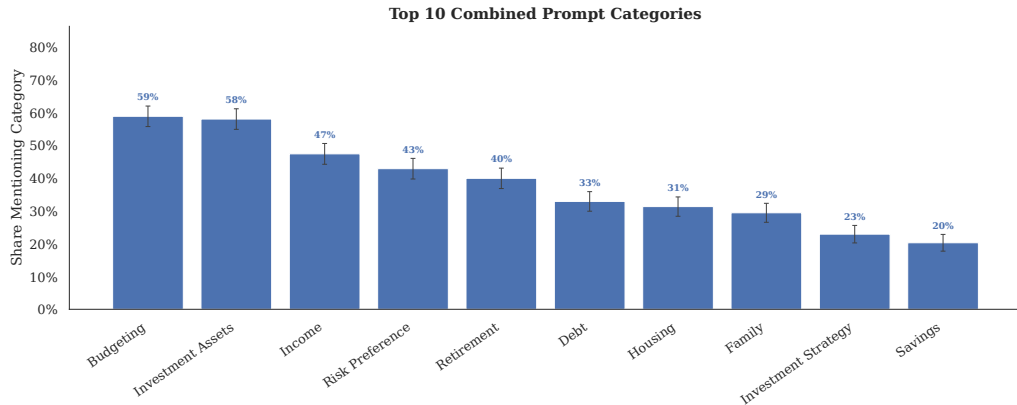
WHAT DO INDIVIDUALS WRITE ABOUT?

Individuals' Prompts

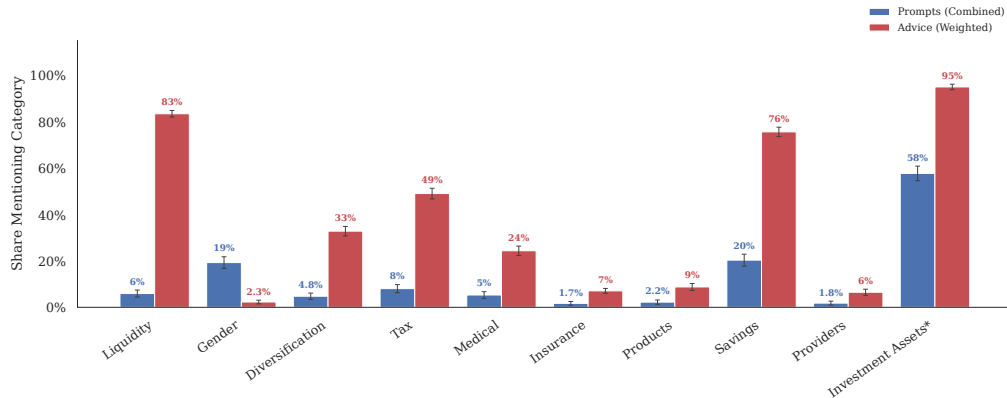


Word Size $\propto \text{Pr}(\text{word})$

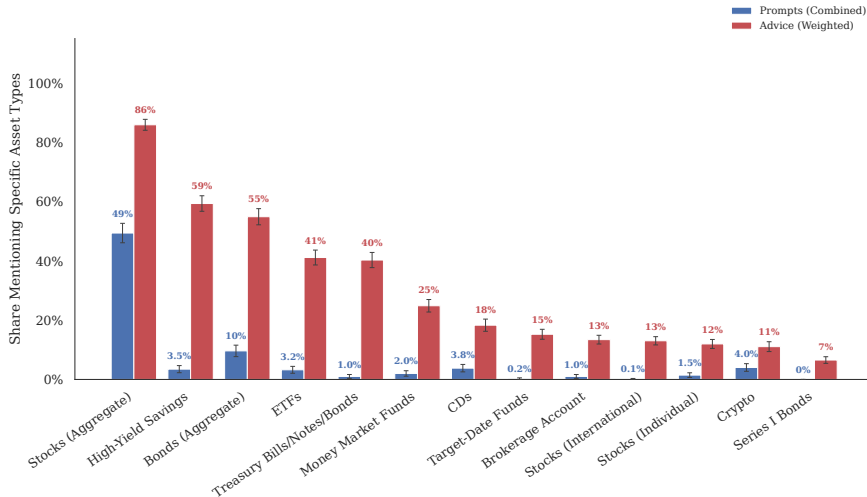
WHICH TOPICS DO INDIVIDUALS DISCUSS?



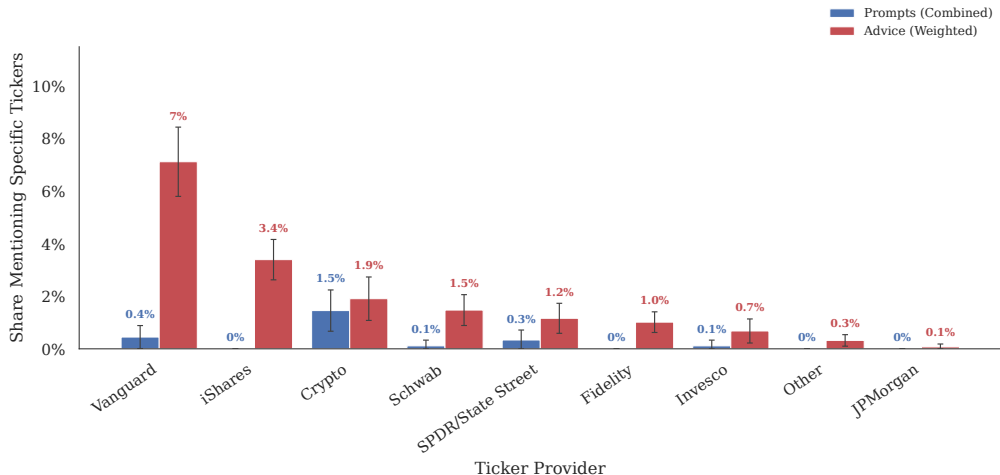
How Do LLM RESPONSES DIFFER FROM PROMPTS?

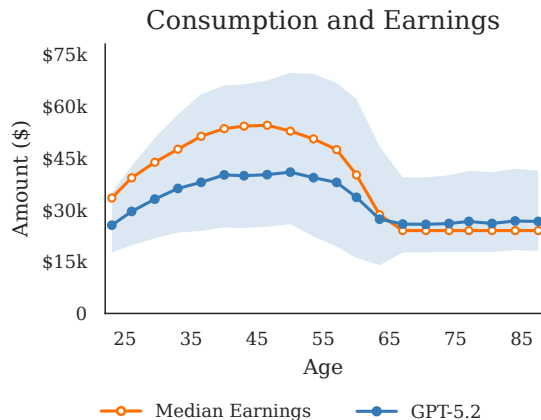


WHICH INVESTMENT ASSET CLASSES ARE DISCUSSED?

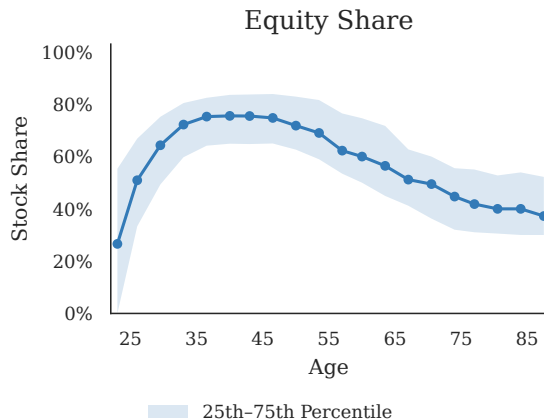
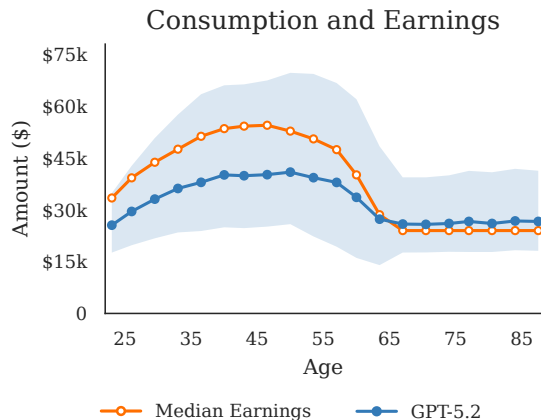


WHICH INVESTMENT PRODUCTS ARE DISCUSSED?

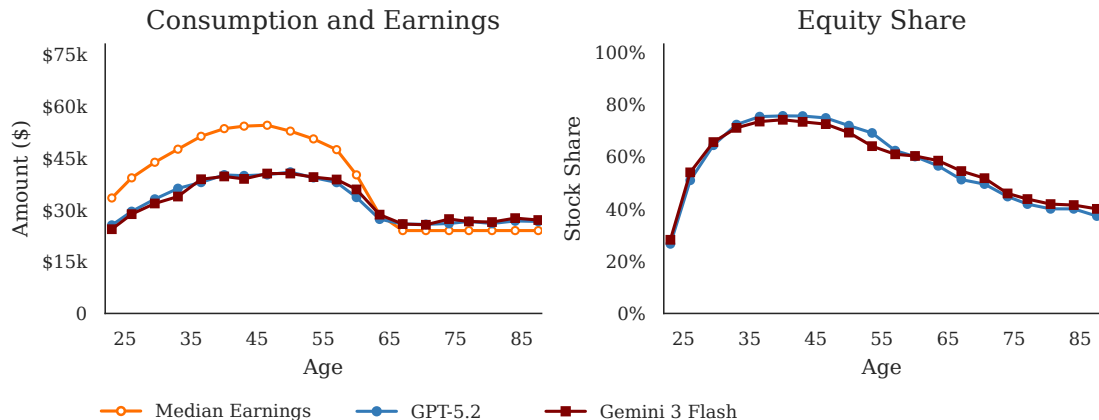




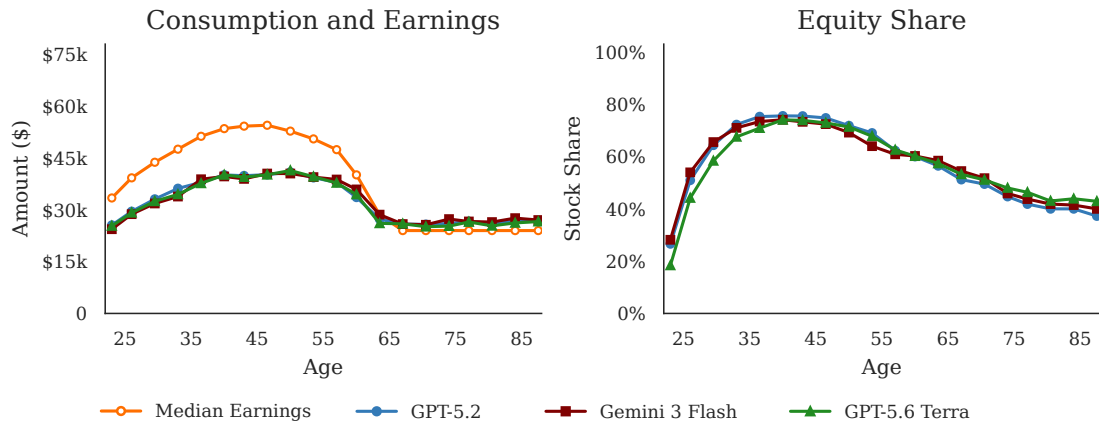
QUANTITATIVE ADVICE: BASELINE SIMULATION



QUANTITATIVE ADVICE: OTHER MODELS



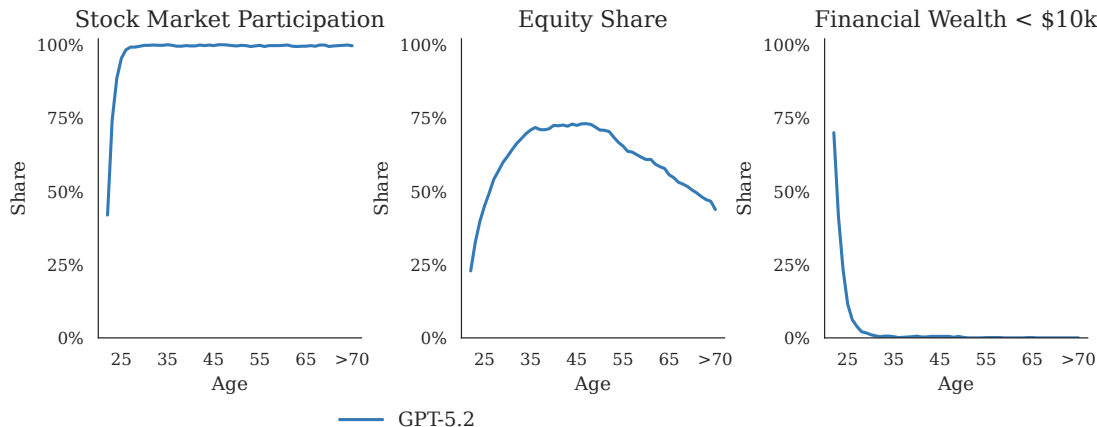
QUANTITATIVE ADVICE: OTHER MODELS



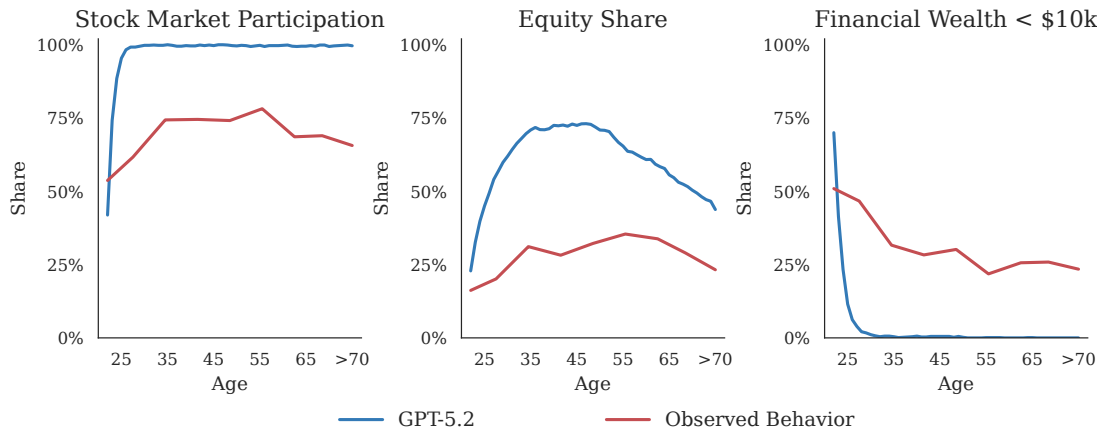
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Fact 1: Advice moves most individuals towards life cycle theory

OBSERVED BEHAVIOR VERSUS LLM ADVICE

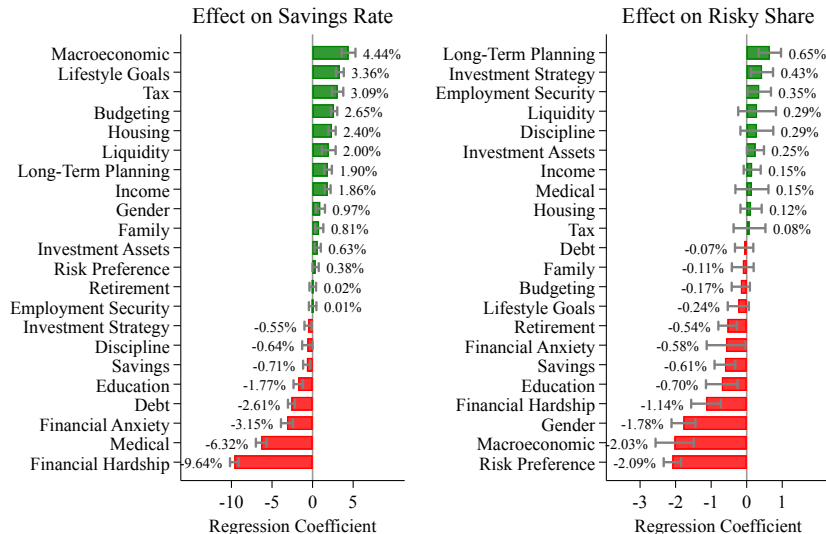


OBSERVED BEHAVIOR VERSUS LLM ADVICE



Higher stock market participation, age-declining equity shares after 45, and larger precautionary buffers

LLM ADVICE RESPONDS SENSIBLY TO PROMPT TOPICS

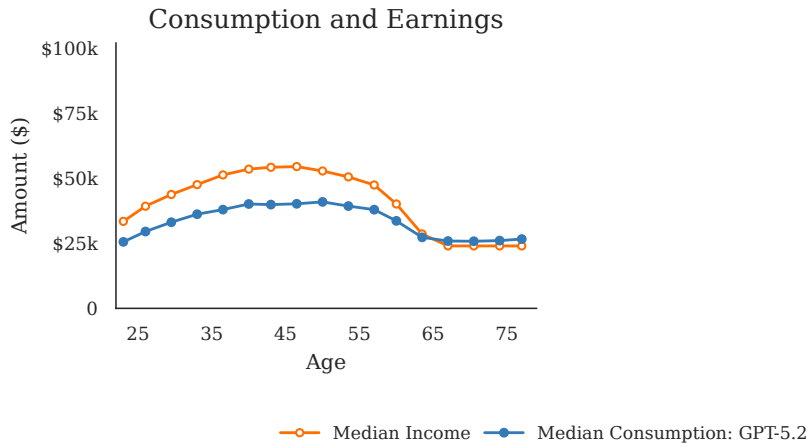


THREE FACTS ABOUT LLM FINANCIAL ADVICE

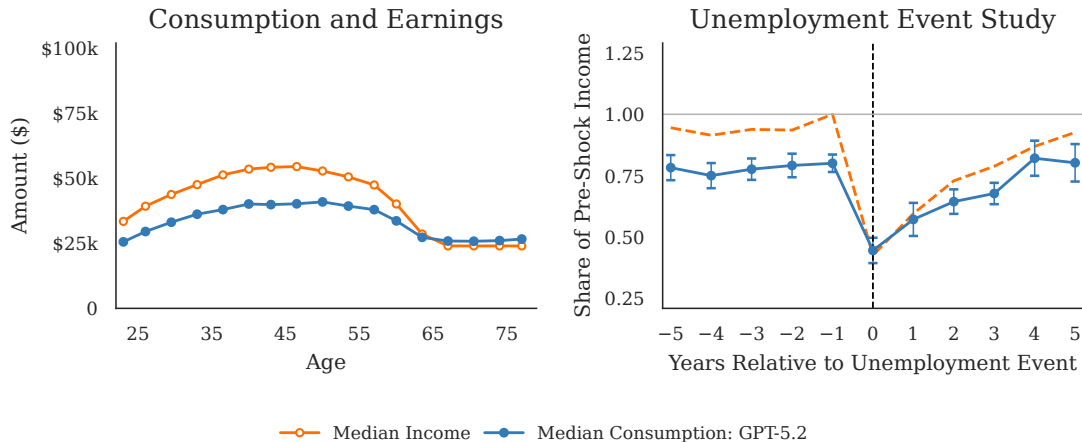
Fact 1: Advice moves most individuals towards life cycle theory

Fact 2: Advice departs from life cycle theory in systematic ways

LLM ADVICE INSUFFICIENTLY SMOOTHS CONSUMPTION

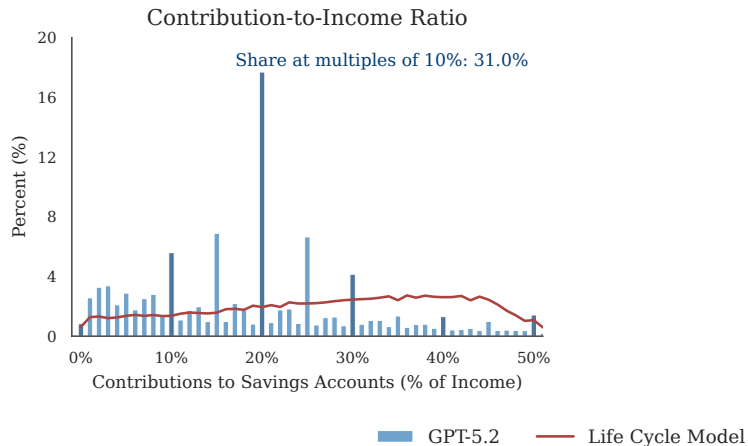


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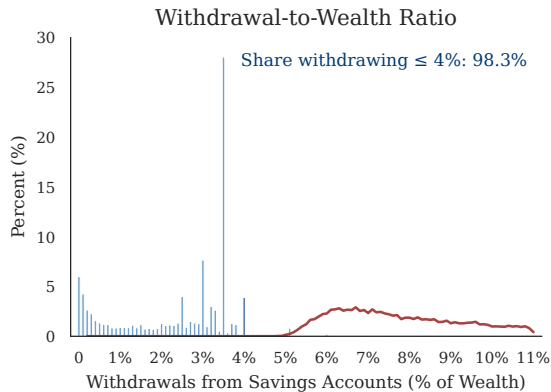
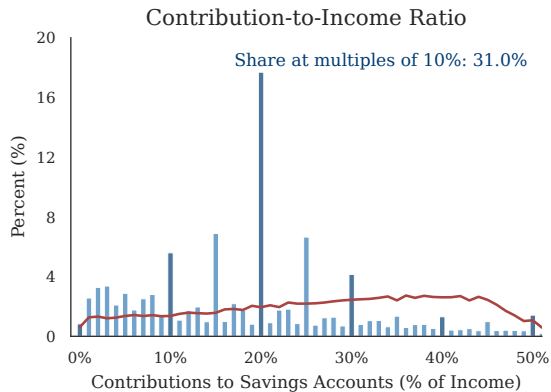
Consumption drops at unemployment despite large buffer stocks (Gruber 1997, Ganong-Noel 2019)

LLM ADVICE RELIES ON HEURISTICS



Savings rates bunch at round numbers...

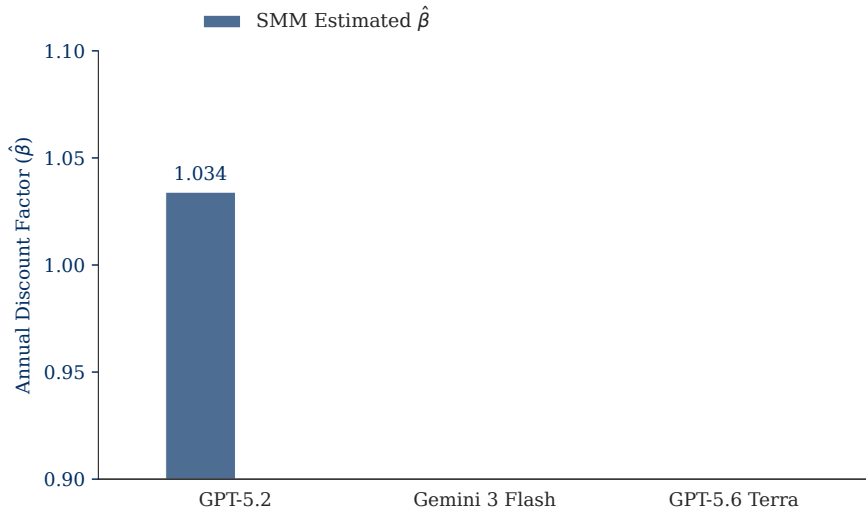
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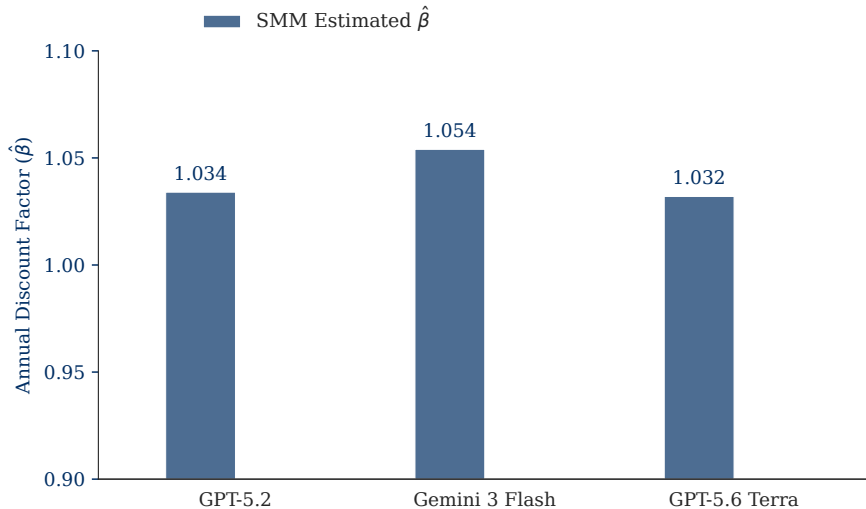
■ GPT-5.2 — Life Cycle Model

... and withdrawals respect the popular 4% withdrawal rule (Cooley et al. 1998)

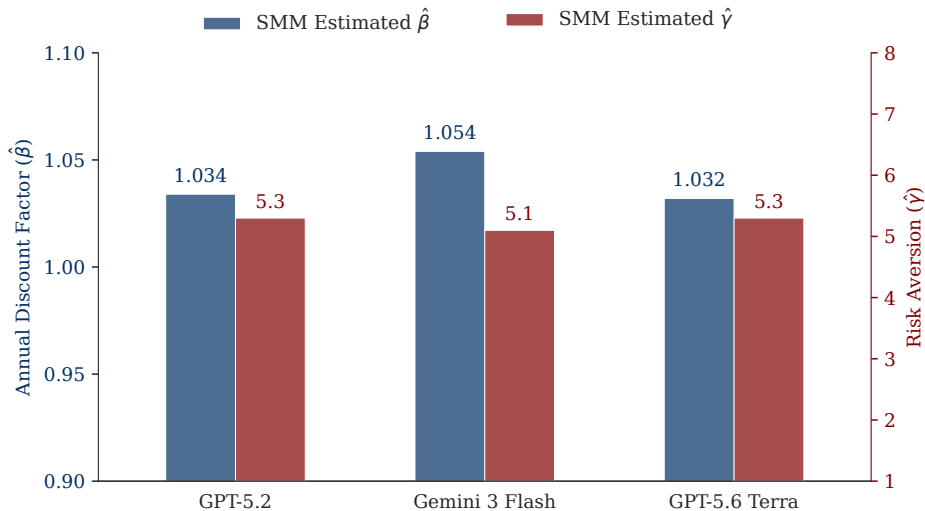
LLM ADVICE IMPLIES UNUSUALLY HIGH PATIENCE



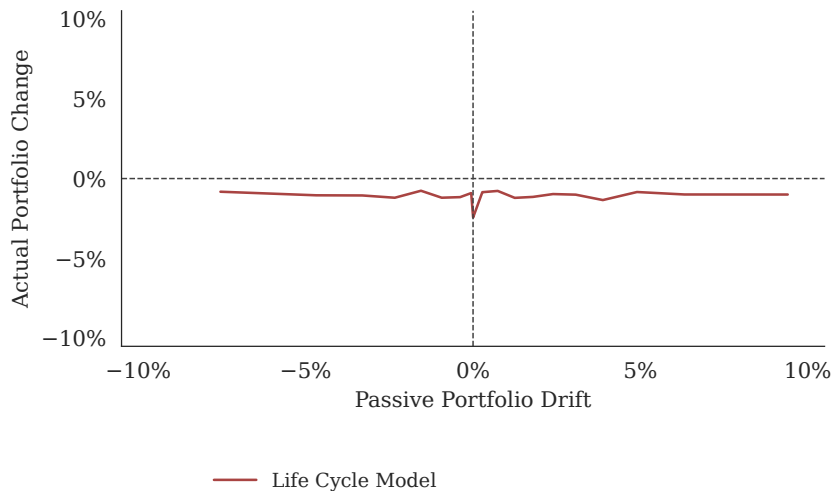
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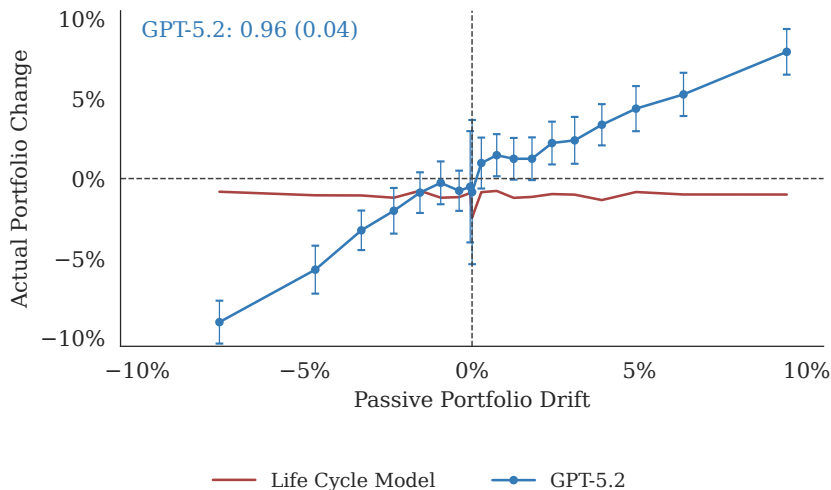
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LLM PORTFOLIOS EXHIBIT SUBSTANTIAL INERTIA



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Equity shares drift passively with realized returns

(Calvet et al. 2009)

ACADEMIC PROMPT REDUCES SOME DEPARTURES FROM THEORY

- ① **Fewer heuristics:** No bunching at round numbers + more withdrawals
- ② **Better consumption-smoothing** over the life cycle and during unemployment
- ③ **More realistic patience:** SMM estimates become $\hat{\beta} = 0.99$ and $\hat{\gamma} = 4.7$
- ④ **No change in portfolio inertia:** Actual–passive portfolio slope unchanged

ACADEMIC PROMPT REDUCES SOME DEPARTURES FROM THEORY

- ① **Fewer heuristics:** No bunching at round numbers + more withdrawals
- ② **Better consumption-smoothing** over the life cycle and during unemployment
- ③ **More realistic patience:** SMM estimates become $\hat{\beta} = 0.99$ and $\hat{\gamma} = 4.7$
- ④ **No change in portfolio inertia:** Actual–passive portfolio slope unchanged

⇒ LLM advice depends on characteristics of **prompts** (demand)

THREE FACTS ABOUT LLM FINANCIAL ADVICE

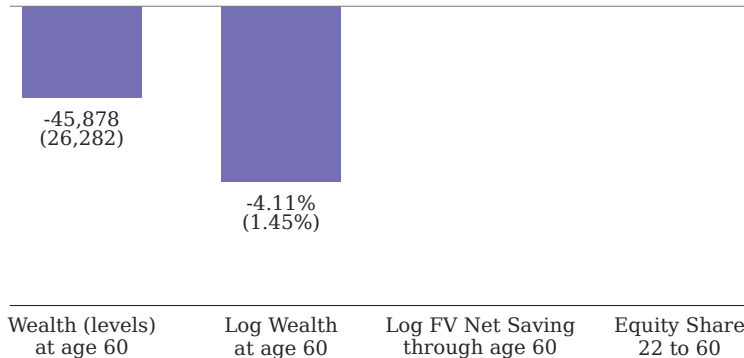
Fact 1: Advice moves most individuals towards life cycle theory

Fact 2: Advice departs from life cycle theory in systematic ways

Fact 3: **Advice varies systematically across groups**

LOW-FINANCIAL-LITERACY PROMPTS ACCUMULATE LESS WEALTH

Financial Literacy: Low vs. High

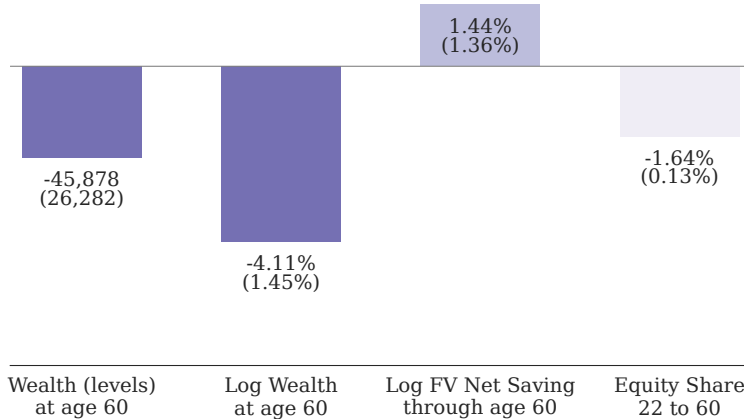


Note: Identical realization of shocks across simulations

(Lusardi and Mitchell 2023)

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High Financial Literacy User:

I am a short term futures trader usually executing **3 to 4 trades per daily session**. I do not hold any positions after the market closes. I **trade the futures micro contracts of indexes, crude oil, gold, currencies and bonds**. I also place option trades on the SPX. I am looking to improve my strategy which is based on Cumulative Delta and Orderflow. My account size is 10,000 USD
I'd like to know how much to spend over the coming year.

I'd like to know how to invest to improve my strategy and reduce my risk exposure.

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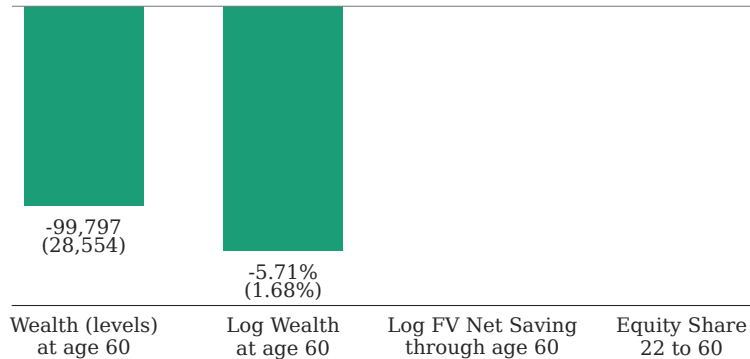
I'd like to know how to invest to improve my strategy and reduce my risk exposure.

Low Financial Literacy User:

Hello....Im in need of some financial advice. I need to invest some money into something that will make me money in the future. [...] I have current stocks invested, however, **I know very little about stocks** and how and when to buy/sell, and **what stocks I need to buy and/or avoid**? Some advice would be very helpful! Thank you! :)

PROMPTS WITHOUT AI EXPERIENCE ACCUMULATE LESS WEALTH

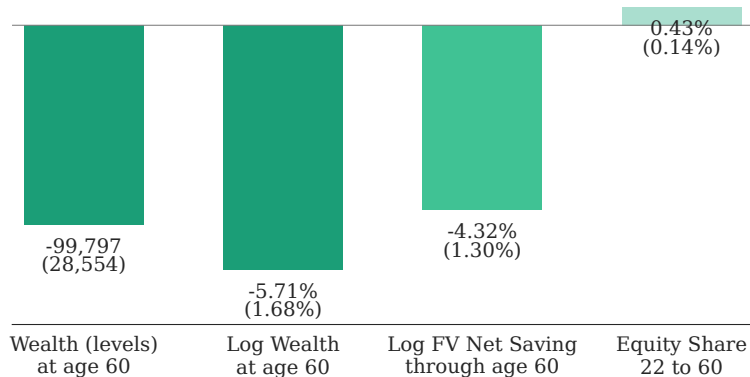
Prior AI Experience: No vs. Yes



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PROMPTS WITHOUT AI EXPERIENCE ACCUMULATE LESS WEALTH

Prior AI Experience: No vs. Yes



Note: Identical realization of shocks across simulations

AI User:

Assume you are a CFA with 20+ years of experience. I am needing advice to help create a sustainable investment portfolio using E*trade or similar self-select investment platforms. I have \$5K to invest currently. I have another \$10k in savings and am already investing in an employer sponsored retirement account contributing the max allowing amount.

Using your CPA knowledge, and our investment strategy determined previously, how much would I be able to continually spend to add to my investment portfolio?

What additional investment vehicles should I be looking add for my further spending? Should reinvest in high return vehicles or expand and diversify?

PRIOR AI EXPERIENCE: EXAMPLE PROMPTS

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AI Non-User:

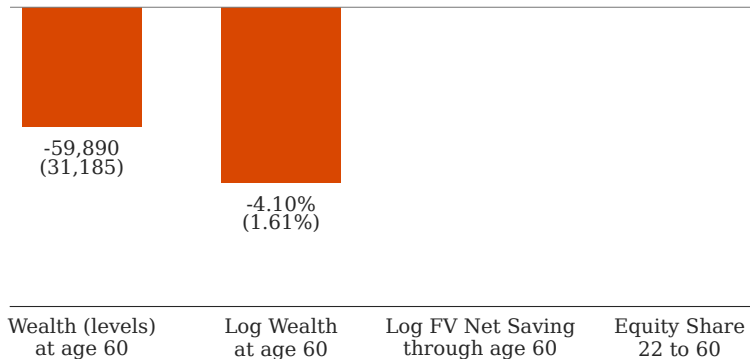
I would like the AI tool to create a budget for my new pay rate as I am starting a new job. I will be making \$20 an hour and working 40-45 hours a week. I pay \$1000 in rent & around \$500 in utilities/ household need bills.

How much could I spend freely with this salary while maintaining funding for rent, household needs, & \$200 savings a month?

Where should I invest starting with \$50 & consistently adding \$25 a month after?

PROMPTS FROM WOMEN ACCUMULATE LESS WEALTH

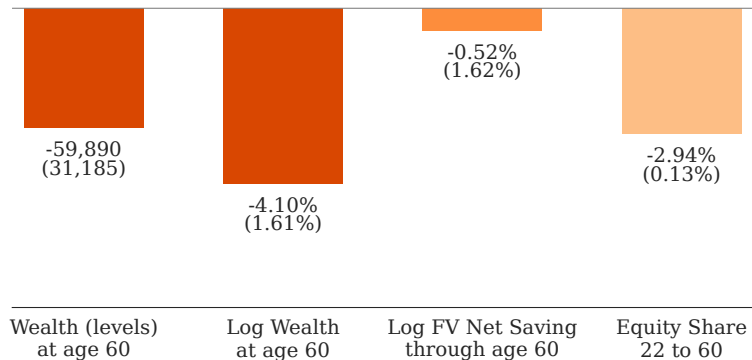
Gender: Female vs. Male



Note: Identical realization of shocks across simulations

PROMPTS FROM WOMEN ACCUMULATE LESS WEALTH

Gender: Female vs. Male



Note: Identical realization of shocks across simulations

- 1 Methodology for Simulating LLM Financial Advice
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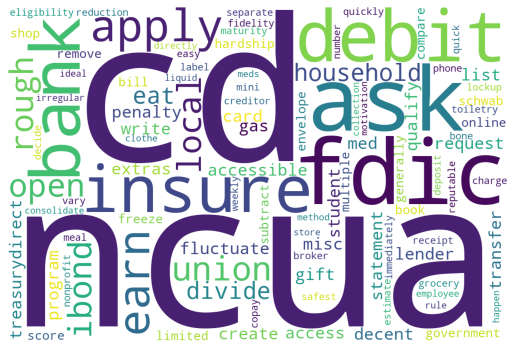
Are these Differences Driven by
Demand (Questions Asked)
or **Supply** (Model Response)?

DIFFERENCES IN LLM RESPONSES BETWEEN MEN AND WOMEN

Men



Women



Word Size $\propto \Pr(\text{word} \mid \text{group}) - \Pr(\text{word} \mid \text{other group})$

Experiment: Consider prompts that do not identify gender (83%) and randomly insert:

“I am a man” or *“I am a woman”*

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- $\text{FemaleAuthor}_i = 1$: prompt *actually* written by a woman
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SEPARATING SUPPLY AND DEMAND: METHODOLOGY

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	Male Label	Female Label
Male Author	0 (reference)	
Female Author		

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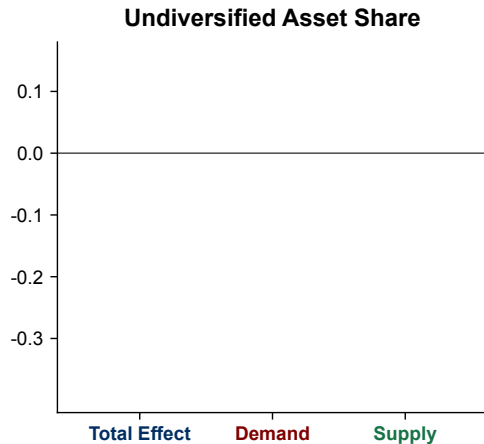
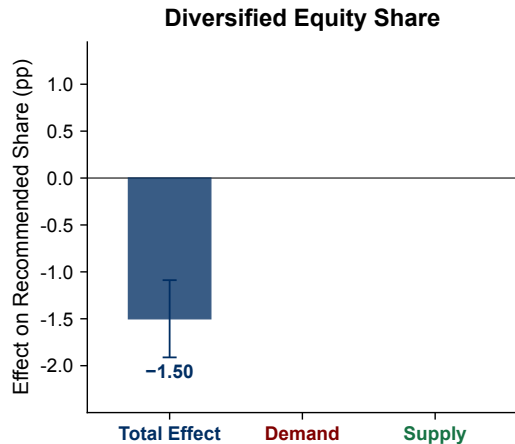
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	Male Label	Female Label
Male Author	0 (reference)	β_S (supply)
Female Author	β_D (demand)	$\beta_D + \beta_S$ (total)

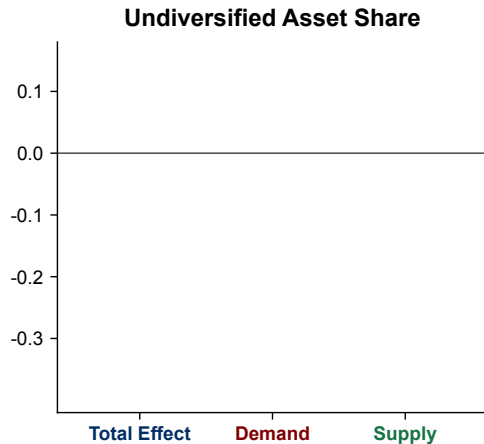
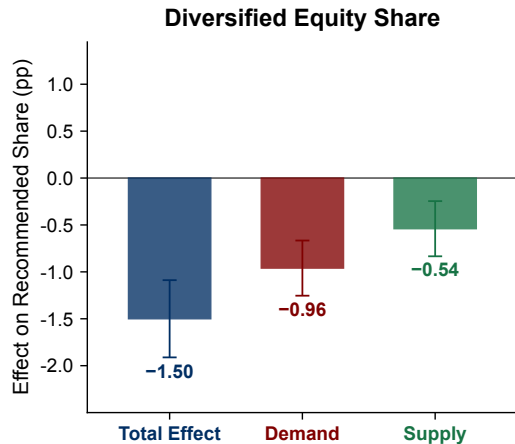
SEPARATING SUPPLY AND DEMAND: FEMALE VERSUS MALE

$$Y_{i,t} = \beta_D \text{FemaleAuthor}_i + \beta_S \text{FemaleLabel}_{i,t} + \delta_{i,t}^{\text{bucket}} + \varepsilon_{i,t}$$



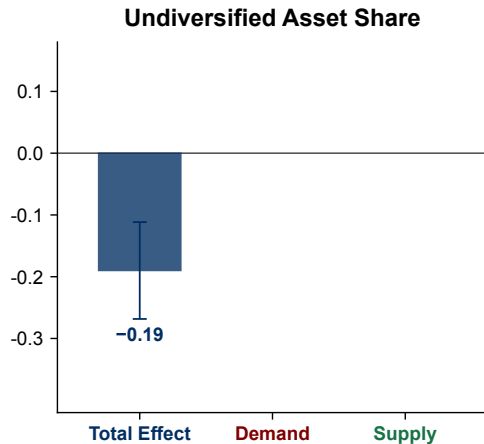
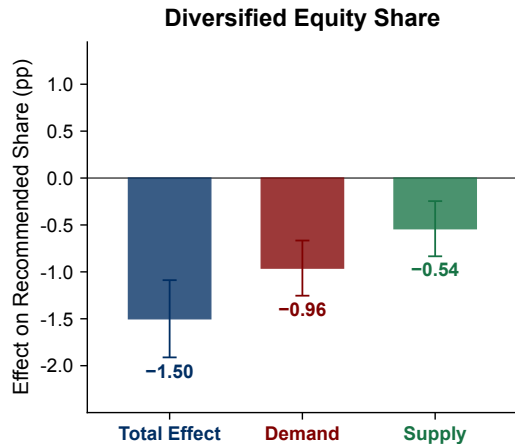
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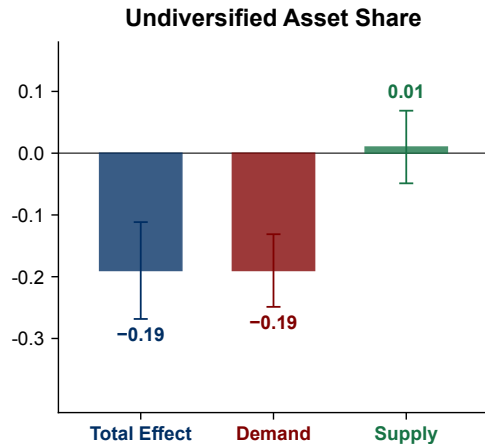
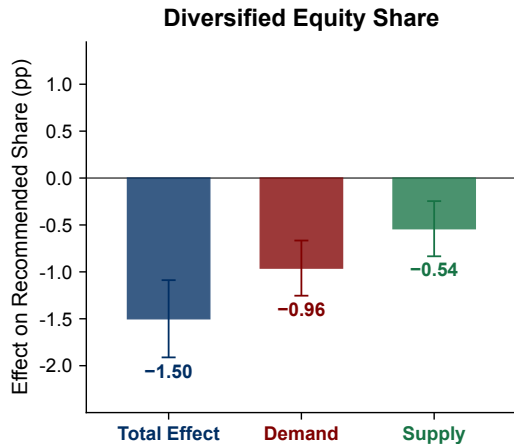
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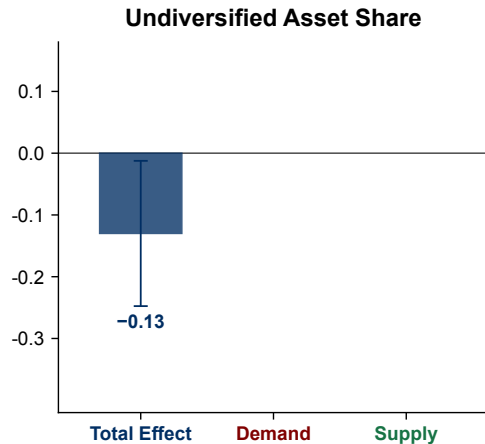
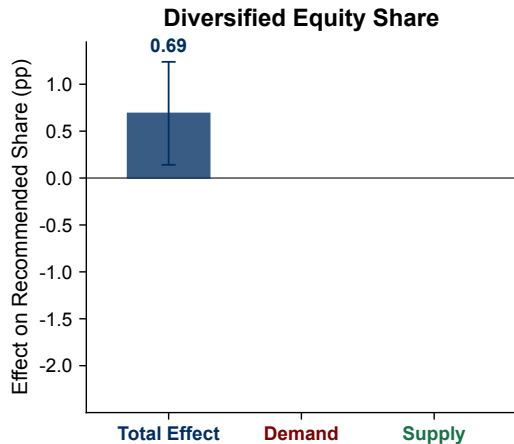
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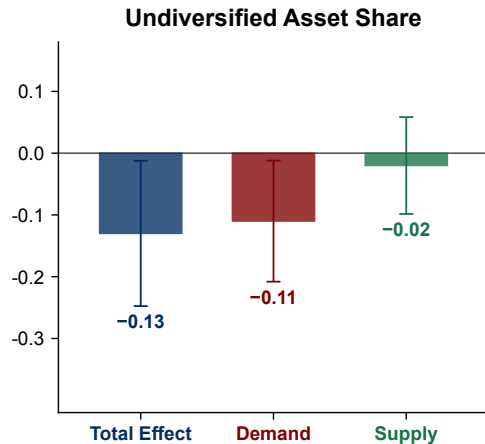
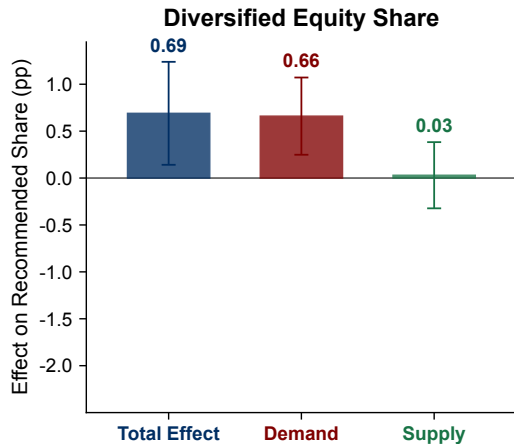
SEPARATING SUPPLY AND DEMAND: BLACK VERSUS WHITE

$$Y_{i,t} = \beta_D \text{BlackAuthor}_i + \beta_S \text{BlackLabel}_{i,t} + \delta_{i,t}^{\text{bucket}} + \varepsilon_{i,t}$$



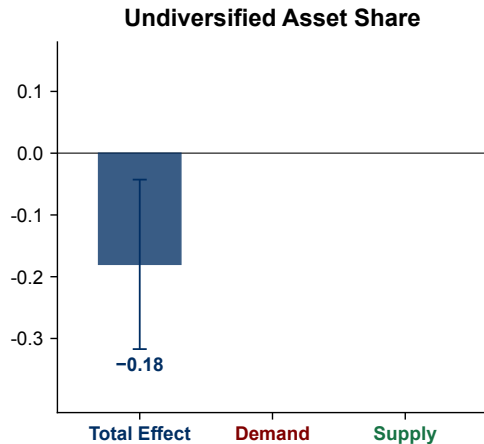
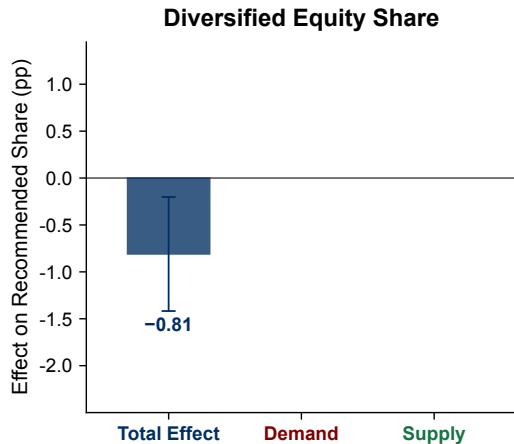
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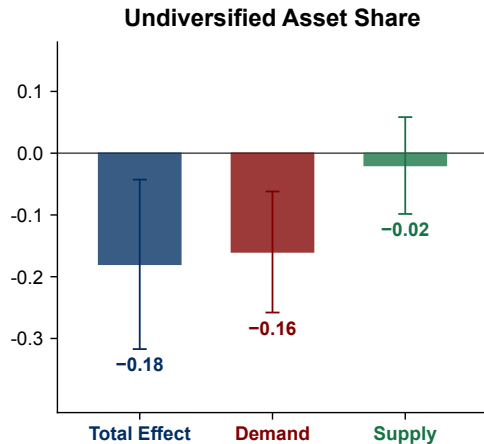
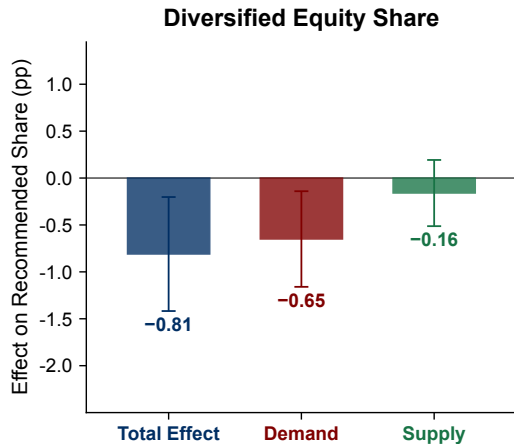
SEPARATING SUPPLY AND DEMAND: ASIAN VERSUS WHITE

$$Y_{i,t} = \beta_D \text{AsianAuthor}_i + \beta_S \text{AsianLabel}_{i,t} + \delta_{i,t}^{\text{bucket}} + \varepsilon_{i,t}$$



SEPARATING SUPPLY AND DEMAND: ASIAN VERSUS WHITE

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Separating **supply** from **demand** is **normatively important**

- **Supply**-driven differences may or may not be desirable
 - **Implicit inference**: demographics correlate with advice-relevant characteristics
 - Or **algorithmic bias**: reproduces training-data stereotypes
- **Supply** shaped by design: no race effect likely reflects **training safeguards**
- **Demand**-driven differences reflect what people actually write about
 - Good for **personalization**, but may amplify existing biases

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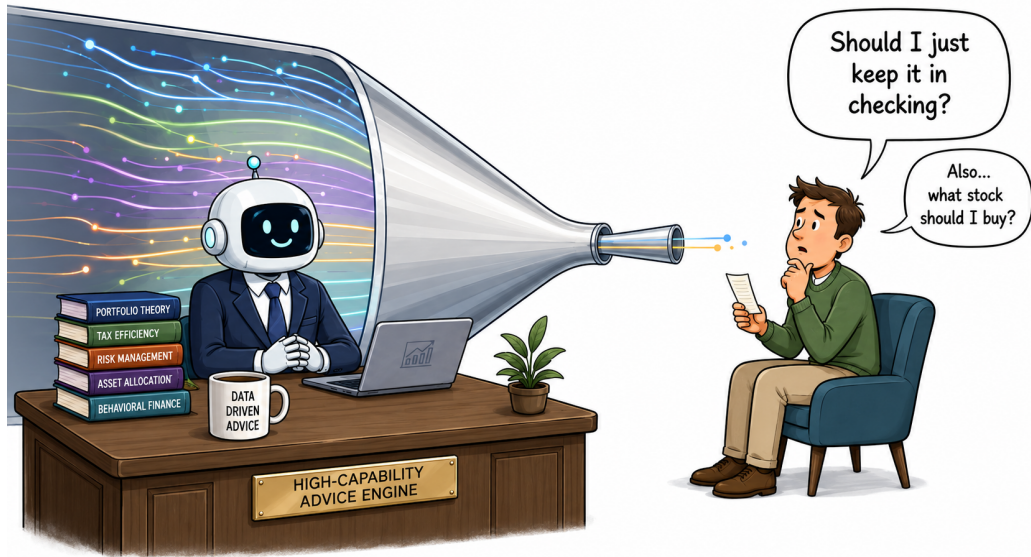
How will these results change as models evolve?

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How will these results change as models evolve?

This paper:

- ① **Scalable methodology** to track model and user evolution
- ② **Diagnostic tests grounded in life cycle theory** that can guide model design
- ③ **Demand-side** will likely evolve more slowly, remaining a binding constraint



Source: ChatGPT-5.6